



APPLICATION NOTE

Measuring CMIS firmware update time with the Binho Supernova

AN0011

Document information

Keywords	AN0011, Supernova, Pulsar, CMIS, OIF, CDB, firmware update, LPL, EPL, tWRITE, tCDBF, tBPC, Nmax, optical module, hold-off
Abstract	Updating firmware on an optical module is slow, and the obvious remedy is a faster management bus. Working through the CMIS access limits and the Table 10-4 timing parameters shows that the wire is not where the time goes: for a 1 MiB image the bus clock accounts for a small fraction of the total and the per-block hold-offs account for most of it. This application note derives the budget from the specification, states plainly which numbers are arithmetic and which would have to be measured, and supplies a command that times the access phases of a real module against the specification ceilings so that a host can find out which hold-offs its module actually takes.

Draft, revision 0.3. The budget in Section 4 is arithmetic derived from the specification. The timing output in Section 5 is a real run, against a **Binho Supernova in I2C target mode**, not an optical module, and it turns out to measure the host path rather than the target. Section 7 says so plainly and states what is outstanding. This revision is for internal review and is not for publication.

1 Introduction

I3C single data rate runs roughly 31 times faster than I2C Fast-mode on the wire. A host engineer facing a slow module firmware update reasonably expects a faster management bus to make it roughly 31 times quicker.

Working the arithmetic through the CMIS access limits gives a different answer, and the difference is actionable: the choice of which Common Data Block command carries the payload matters more than the choice of bus.

This note derives the budget, and supplies a way to find out where a specific module actually spends its time rather than where the specification permits it to.

This document does not state a measured throughput. Section 7 explains why, and what would have to be run to state one.

1.1 Scope

The document covers the arithmetic of a CMIS firmware download: the access length limits, the block sizes of the two Write Firmware Block commands, and the Table 10-4 hold-off parameters. It supplies a command that times the read, page change and Common Data Block access phases of a module against those ceilings.

It does **not** write firmware to a module, and the supplied utility has no command that does. It does not cover the firmware update state machine, image signing, or the Run and Commit sequence beyond noting their timing. AN0008 covers the register model and is a prerequisite.

1.2 Applicable devices

Table 1. Applicable devices

Item	Role	Basis
Binho Supernova	Host adapter	Exercised, Section 7
Binho Pulsar	Host adapter	Not exercised
Any CMIS managed module	Target	Not exercised. The measurement used an adapter in target mode

The budget in Section 4 is arithmetic, not measurement. It is derived from OIF-CMIS-05.3 sections 5.2.2 and chapter 9 and from Table 10-4, and is reproducible with `python cmis_mci.py budget`. No row of it has been checked against a module.

1.3

Related documents

Table 2. Related documents

Document	Subject
OIF-CMIS-05.3	Sections 5.2.2.1 and 5.2.2.2, chapter 9 for the CDB, Table 10-4 for timing
AN0008	Bringing up a CMIS optical module over I2C with the Binho Supernova
AN0009	Managing a CMIS optical module over I3C with the Binho Supernova
AN0010	Bringing up a CMIS optical module over SPI with the Binho Pulsar

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Required equipment and software

Table 3. Required equipment

Item
Binho Supernova or Binho Pulsar USB host adapter, for the <code>timing</code> command
A CMIS managed optical module, for the <code>timing</code> command
Nothing at all, for the <code>budget</code> command

```
pip install binhosupernova
python cmis_mci.py budget
```

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The four facts the budget rests on

3.1

A firmware block over the Local Payload is 116 bytes

The Common Data Block occupies Page 9Fh. Its Local Payload is 120 bytes, of which `Write Firmware Block LPL`, command `0103h`, spends 4 on the block offset. That leaves **116 bytes** of firmware per block.

The Extended Payload is pages A0h through AFh, 16 pages of 128 bytes, giving **2048 bytes** per block through `Write Firmware Block EPL`, command `0104h`.

3.2 A WRITE carries at most 8 bytes

Section 5.2.2.2 states it directly: "A successful WRITE writes a sequence of up to eight given byte values". `FullPageReadSupported` raises `Nmax` to 128 for **reads**, and does nothing for writes.

The only escape is the note immediately following: "A WRITE with more than 8 bytes may be allowed when specified explicitly in chapter 7.8", which is the Common Data Block Extended Payload region.

So a 116 byte Local Payload block is $\text{ceil}(116 / 8) = 15$ **separate write transactions**, each with its own address phase.

3.3 The module then stops talking

Table 4. Hold-off parameters, CMIS Table 10-4 maxima

Parameter	Maximum	Applies after
tREAD	500 microseconds	A register read
tWRITE	10 ms	A write to volatile memory
tWRITENV	80 ms	A write to non-volatile memory
tBPC	10 ms	A bank or page change
tCDBF	4960 ms	A foreground CDB command. The default is 80 ms
tMgmtInit	2000 ms	Reset, until ModuleLowPwr is reached

These are maxima. A module may advertise anything up to them, and Section 5 is about finding out what a given module actually takes.

3.4 Almost all CMIS memory is volatile

The specification states it outright, so **tWRITE at 10 ms is the common case**, not the optimistic one.

The hold-off is paid **once per block**, not once per transaction. That is the whole lever: a 2048 byte block pays it 512 times over a 1 MiB image, and a 116 byte block pays it 9040 times.

4 The budget

```
python cmis_mci.py budget --image-kib 1024
```

Firmware update budget, 1024 KiB image, tWRITE 10 ms per block

path	blocks	per block	bus	hold-off	total
I2CMCI 400 kHz, LPL 0103h	9040	15 txn	36.61 s	90.40 s	127.01 s
I3C SDR 12.5 Mb/s, LPL 0103h	9040	15 txn	1.17 s	90.40 s	91.57 s
I2CMCI 400 kHz, EPL 0104h	512	16 txn	24.33 s	5.12 s	29.45 s
I3C SDR 12.5 Mb/s, EPL 0104h	512	16 txn	0.78 s	5.12 s	5.90 s

Bus clock alone, 400 kHz to 12.5 Mb/s, about 31 times the wire speed: 1.39 times faster.

Payload size alone, LPL to EPL at 400 kHz: 4.3 times faster.

Both together: 21.5 times faster.

Read the first two rows against each other. Multiplying the wire speed by 31 removes 35 seconds. Changing which Common Data Block command carries the payload, at the original slow clock, removes 97.

Table 5. What the budget model does and does not include

Included	Excluded
Transactions per block, from the access length limits	Host USB latency, which is not a bus property
Bus time per transaction, as 4 framing bytes plus payload at 9 bits each	tCDBF, which varies by module and command
tWRITE once per block	The Run Image and Commit Image steps
	Retries after a rejected access

The bus time column is an estimate of wire occupancy, not a measured throughput. It is deliberately crude, because its only job is to show that it is small next to the hold-off column. A more careful model of it would not change the conclusion, and a measured value would characterize the host's USB path as much as the bus.

5 Measuring a real module

```
python cmis_mci.py timing
```

The command writes no firmware. It times the access phases an update is built from, a single byte read, an eight byte read at the write limit, a full page read where the module advertises one, a page change in both directions, and reaching the Common Data Block page, and prints each next to its Table 10-4 ceiling.

```
$ python cmis_mci.py timing
```

phase	median	max	CMIS max
single byte read	2.058 ms	3.400 ms	0.5 ms
8 byte read, the WRITE limit	1.995 ms	2.064 ms	-
page change 00h to 01h to 00h	7.988 ms	9.019 ms	20.0 ms
select CDB page 9Fh	4.043 ms	5.913 ms	10.0 ms

One 116 byte LPL firmware block is 15 write accesses, about 29.9 ms of bus time, against a tWRITE hold-off of up to 10 ms that the module pays once per block.

Read that as a measurement of the host, not of the target. Every row is roughly two milliseconds per USB round trip: one access costs about 2 ms, the two register writes of a CDB page selection cost about 4, and the four writes of a page change in both directions cost about 8. A single byte read sits four times above tREAD, and an eight byte read is not slower than a one byte read, which is only possible if the bytes on the wire are not what is being timed.

The target in this run is an adapter in I2C target mode, which answers immediately and never holds the bus off. So what the numbers show is the floor the host path imposes before a module has contributed anything at all, and that floor is around 2 ms per access on this setup. Section 7.2 draws the conclusion.

What to look for in the result:

A median far below the specification maximum means the module is faster than it is permitted to be, and a host that waits the maximum after every write is leaving most of the update time unclaimed. This is the most valuable thing the measurement can tell you.

A maximum close to the specification ceiling on a page change means the module really does take tBPC sometimes. A host that assumes the median will fail intermittently, which is the worst failure mode to debug.

A single byte read above tREAD is host side or adapter side overhead, not module hold-off. tREAD is an abstraction over I2C clock stretching, and clock stretching is visible on a bus capture in a way that a host side timer cannot distinguish from anything else.

6 What to do with the finding

For host software. Query `Firmware Management Features`, CDB command `0041h`, once at the start of a download and prefer `Write Firmware Block EPL (0104h)` over `0103h` where the module supports it. Do not assume either way.

For module firmware. Advertise hold-off times you can honor. `tWRITE` is a maximum, not a target, and the measurement in Section 5 is how a host discovers what you left unclaimed.

For anyone specifying a new MCI binding. SPIMCI already carries up to 2048 bytes in one transaction, and I3C negotiates the same thing natively through SETMWL, SETMRL and GETMXDS. A binding that inherits I2CMCI's 8 byte write limit inherits this entire problem with it.

Budget the Run Image step separately. CDB command `0109h` takes as long as a full module reboot, "mainly governed by the `tMgmtInit` timing parameter", so allow roughly two seconds for it rather than a CDB timeout.

7 Measured results

7.1 Test configuration

Table 6. Test configuration

Item	Value
Controller	Binho Supernova, serial <code>1912DEB5...79FA</code> , firmware 4.4.0, hardware revision C
Target	Binho Supernova, serial <code>ED7BF8F3...F7BB</code> , firmware 4.4.0, armed as an I2C target at 0x50
Target contents	CMIS 5.3 image, the Appendix C 400G QSFP-DD worked example
SDK	<code>binhosupernova</code> 4.2.0, Python 3.10.12
Host	Ubuntu 22.04, x86-64
Bus	3.3 V, 400 kHz
Repeats	20 per phase, median and maximum reported

7.2 Results

Derived. The budget in Section 4 is reproducible from the supplied utility and rests only on the specification citations in Section 3. It is arithmetic, and this document does not present it as anything else.

Measured, and it measures the host. The Section 5 run establishes a floor of about 2 ms per register access on this host, adapter and SDK, with a target that imposes no hold-off of its own. That number belongs to the USB path, not to the bus and not to a module: a single byte read exceeds `tREAD` fourfold, and an eight byte read is no slower than a one byte read, both of which are only possible if wire time is not the term that dominates.

This is the reason **no end to end throughput figure appears anywhere in this document**. A figure measured this way would characterize the host, and quoting it as a bus or module

property would be wrong in a way that is hard to detect later.

It does, however, sharpen the note's own argument rather than weakening it. Section 4 argues that wire time is not where a firmware update goes. The measurement adds that host round trip cost is not where it goes either. Both are small next to a per-block hold-off, and both shrink for the same reason when a block gets larger: **fewer accesses and fewer blocks**. A 2 ms per access floor costs 30 ms per Local Payload block and 32 ms per Extended Payload block, but the Extended Payload block carries 17 times more firmware.

On hardware, against a module. Nothing here. Outstanding at revision 0.3: the `timing` command against a real module, which is the only thing that establishes the hold-offs a module actually takes, and a complete firmware download over both the Local Payload and Extended Payload paths, which is the only thing that would support an end to end figure.

Until that has been run, the claim this note supports is about **where the time is permitted to go**, and about what the host costs before a module contributes anything.

8 Troubleshooting

Table 7. Troubleshooting

Symptom	Probable cause and action
<code>timing</code> reports a single byte read far above tREAD	Host or adapter overhead, not the module. Compare against a bus capture
Every phase is close to its ceiling	The module may be genuinely slow, or another controller is contending for the bus
The 94 byte read is absent from the output	The module does not advertise full page read, so <code>Nmax</code> stayed at 8
Medians vary widely between runs	Something else is using the USB path. Close other host tools and repeat
A page change exceeds twice tBPC	The module is taking longer than the specification permits. Capture the bus before blaming the host

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Acronyms

Table 8. Acronyms

Term	Meaning
CDB	Common Data Block
CMIS	Common Management Interface Specification
EPL	Extended Payload, CDB pages A0h to AFh
LPL	Local Payload, in CDB page 9Fh
MCI	Management Communication Interface
Nmax	Maximum number of bytes in one register access
OIF	Optical Internetworking Forum
SDR	Single data rate

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Note about the source code in the document

Example code shown in this document and supplied in the accompanying archive is provided for evaluation and reference. It is released under the MIT license, the text of which is included in the archive.

Table 9. Contents of the assets archive

File	Description
<code>AN0011.md</code>	Markdown source of this document
<code>assets/cmisi_mci.py</code>	Reference utility, shared by AN0008 through AN0011
<code>assets/LICENSE</code>	MIT license covering the example code

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Revision history

Table 10. Revision history

Revision	Date	Description
0.1	1 September 2026	Draft for internal review. Budget derived from the specification; no measurement taken.
0.2	1 September 2026	<code>timing</code> run against a Supernova I2C target fixture. Output pasted, and reported as a measurement of the host path rather than of a module.
0.3	2 September 2026	Renumbered from AN0010. No change to the measurements or the budget.

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